|INCH-POUND|

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1 December 1993
SUPERSEDING
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7 June 1993

MILITARY SPECIFICATION SHEET

FILTER ELEMENT, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 3000 PSI, 5 MICRON ABSOLUTE, -65° TO +275°F, F-14 AND F-18 AIRCRAFT, 84 GPM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-8815

REQUIREMENTS:

Envelope. See Figure 1.

Pressure drop. Maximum pressure drop at rated flow (84 GPM) across the filter element shall not exceed 20 psid at 190 ±10°F. The maximum pressure drop at 42 GPM shall not exceed 20 psid at 100 ±10°F.

Contaminants. See Table I.

Degree of filtration. The filter element shall remove 100 percent of all contaminants larger than 5 microns and 95 percent of all test contaminants. The filter efficiency shall be a minimum of 95 percent.

Element collapse. The filter element and element end cap shall withstand a differential pressure of 5500 psid. In addition, the end cap shall withstand a differential pressure at 4500 psid applied in the reverse direction (inside to outside).

Cold start. The element shall withstand 3000 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop following the cold start test shall meet the requirement for pressure drop as shown above.

Maximum particle passed. The maximum particle passed shall be not greater than 5 microns.

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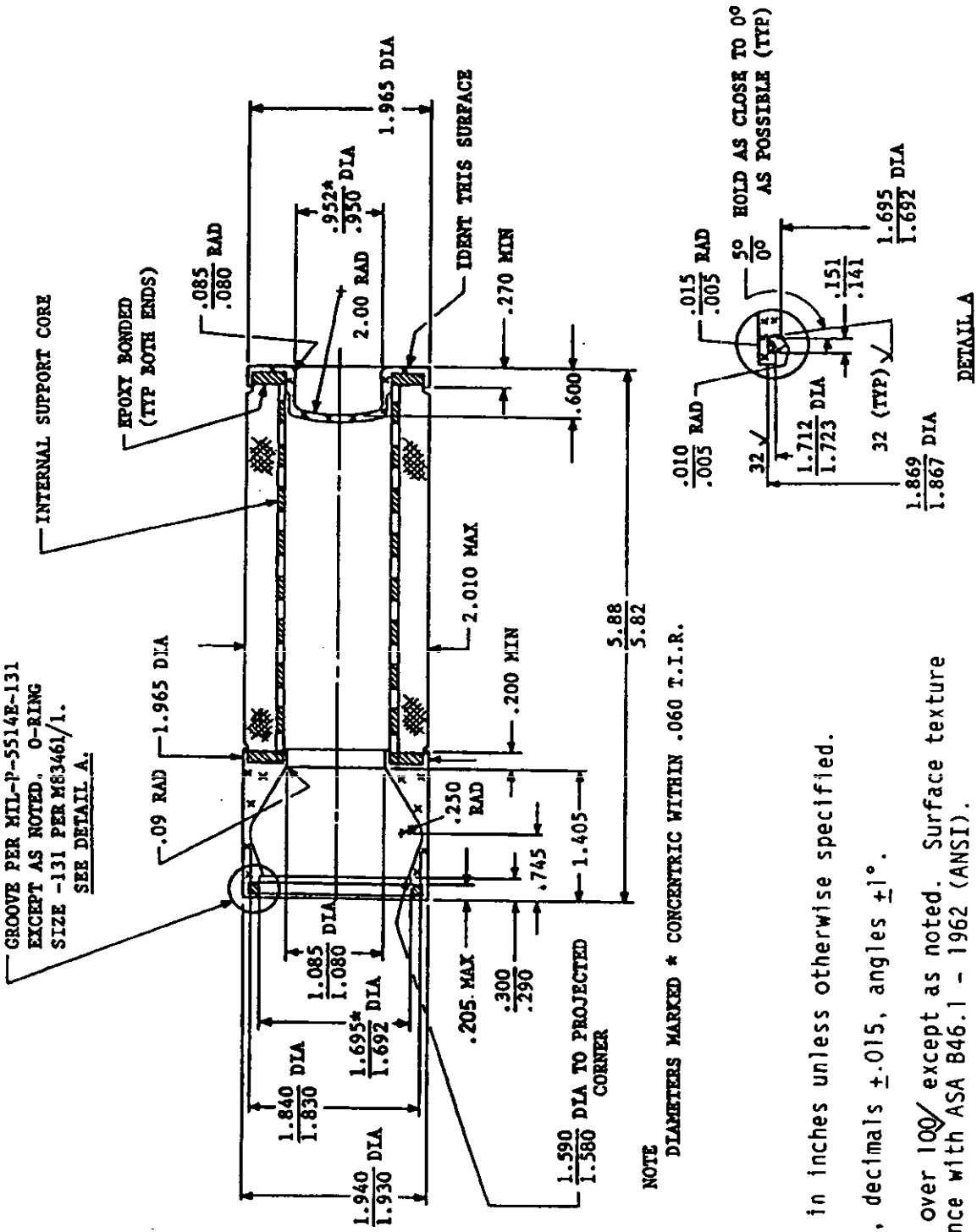


FIGURE 1. Dimensions.

NOTES:

1. Dimensions in inches unless otherwise specified.
2. Tolerances, decimals ±.015, angles ±1°.
3. Finish all over 100 except as noted. Surface texture in accordance with ASA B46.1 - 1962 (ANSI).
4. Material shall be in accordance with the procurement specification.
5. Dome on filter bowl which supports bottom element end cap should have pressure relief drilled hole or chamfer provisions as end cap is rated at 4500 psid.

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TABLE I. Contaminants.

Test Flow Rate (GPM)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup (at 100 ±10°F)		Migration Analysis Value Maximum Allowable (gms)	
	F-9 Beads (gms)	AC Dust (gms)	F-9 Beads (gms)	Carbonyl Iron E (gms)	Add Rate AC Dust (gms)	Minimum AC Dust to 90 psid (gms)	After Cold Start 4.7.2.3	After Vibration 4.7.2.8.3.2
30	0.258	0.258	0.52	0.52	--	---	0.0405	0.0081
42	--	--	--	--	0.50	5.0	---	---
84	--	--	--	--	0.50	3.4	---	---

Flow fatigue. The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cpm maximum.

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
500 cycles at 5000 psid pressure drop

Part number. The part number shall consist of the basic number of this specification sheet.

Example:

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Maximum weight. The maximum weight of the filter element shall be 0.82 pounds.

Intended use. The filter element covered by this specification sheet is intended for use as specified in paragraph 6.1 of the general specification.

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TESTS:

The following tests shall be conducted on filter element number 1 in addition to the tests specified in the general specification. (See Table I of the general specification).

MIL-H-83282 immersion. Following the immersion test specified in the general specification, the filter element shall be immersed in hydraulic fluid conforming to MIL-H-83282 for a period of 72 hours at a fluid temperature of 275°F, and 72 hours at a fluid temperature of -65°F. All internal parts shall be in contact with the test fluid during this period. After immersion, the filter shall remain in this fluid at room temperature until further tests are conducted.

Element bonding test. Following the flow fatigue test specified in Table I of the general specification, the bonding test shall be conducted to verify element assembly integrity and to determine if degradation occured during prior testing. Support the element end cap (closed end) by means of a rigidly mounted fixture. The fixture shall distribute the load around the element end cap periphery. Apply an axial tensile load to the header (open end) for 15 seconds and then gradually reduce the load to zero. The load shall be 25 +5, -0 pounds. Repeat the load cycle five times. After five cycles inspect the element. There shall be no element deformation or bonding failures.

Element end cap integrity test. The outlet port of the filter assembly shall be plugged, and 4500 psi shall be applied at least twice and held for two minutes at each application. The pressure shall be reduced to zero between applications. The filter element shall be removed from the housing and examined for any structural damage. There shall be no evidence of permanent deformation, malfunction, or external leakage.

Preparing Activity:

Navy - AS

(Project No. 1650-N176)

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